Chapter 3 — Chemical Bonding

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Sigma bond forms by: (a) Head-on overlap (b) Electron transfer (c) Sideways overlap (d) No overlap

2. Pi bond forms by: (a) Proton sharing (b) s-s overlap only (c) Head-on overlap (d) Sideways overlap

3. A double bond has: (a) 2 pi (b) 1 sigma + 1 pi (c) 2 sigma (d) 1 pi only

4. sp^3 bond angle is: (a) 90° (b) 120° (c) 109.5° (d) 180°

5. sp^2 shape is: (a) Tetrahedral (b) Bent (c) Linear (d) Trigonal planar

6. NH_3 shape is: (a) Pyramidal (b) Planar (c) Linear (d) Octahedral

7. Bond order of O_2: (a) 0 (b) 1 (c) 2 (d) 3

8. O_2 is paramagnetic due to: (a) Ionic bonds (b) H bonds (c) Unpaired electrons (d) No unpaired e^-

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is a sigma bond stronger than a pi bond?

2. How does sp^3 hybridization explain the shape of methane?

3. Why is the bond angle in water smaller than in ammonia?

4. How does VSEPR theory predict molecular shape?

5. Why is oxygen paramagnetic according to MOT?

6. How does bond order relate to bond strength?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain valence bond theory, sigma/pi bonds and sp, sp^2, sp^3 hybridization. **(4)**

(b) Use VSEPR theory to predict shapes and bond angles of molecules. **(4)**

Q.4 (a) Explain molecular orbital theory and bond order. **(4)**

(b) Use MOT to explain the paramagnetism of oxygen. **(4)**

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